

Bryan C. Busby

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EDUCATION

B.Sc. in Pure Mathematics, University of Puerto Rico

Expected 2027

RESEARCH EXPERIENCE

Villanova University (REU) – Advisor: Alexander Díaz-López

Summer 2025

Collaborated with other undergraduates. We studied arithmetical structures on graphs. In particular, structures on cycles with multi-edges. Manuscript in preparation.

UPR – Advisor: Omar Colón-Reyes

Spring 2024

Research on transient length bounds for Boolean monomial systems using graph-theoretical tools (walks, Frobenius number). Manuscript in preparation.

UPR (CARSE) – Advisor: Rafael Rodríguez Solís

2023

Analyzed instrument tolerance effects on 5G propagation measurements. Learned how difficult debugging software to hardware communication can be on GNU Radio.

High School – Advisor: Edwin Benvenutti

2021

Independent research on the powers of adjacency matrices and its relation to walks along graphs.

MANUSCRIPTS IN PREPARATION

Arithmetical Structures on Cycles with a multi-edge (with Alexander Díaz-López)

Transient Length Boundary for Finite Dynamical Systems (with Omar Colón-Reyes)

PRESENTATIONS

Arithmetical Structures on Cycles with a multi-edge – MAA MathFest, Sacramento, CA

Aug 2025

Finite Dynamical Systems: Transient Bounds for Boolean Monomial Systems – SIDIM
(PR Interdisciplinary Mathematics Symposium), UPR Ponce

May 2024

Development of Radio Wave Propagation Setup and Propagation Measurements at 5G Bands – UPR Mayagüez

2023

On the Adjacency Matrix and Walks – Regional Math Fair (PR Dept. of Education)

2021

TEACHING EXPERIENCE

Private Mathematics Tutor

2022–present

Provided one-on-one tutoring at the high school and early undergraduate level. Accumulated over 100 hours of tutoring experience.

EMPLOYMENT

Food Server, Resort & Casino, PR

2025–present

Food Server, Romano's Macaroni Grill, PR

2023–2024

LANGUAGES

Spanish (native); English (fluent).